

BAAZ & UDAY JKBOSE

2026 CLASS 11TH



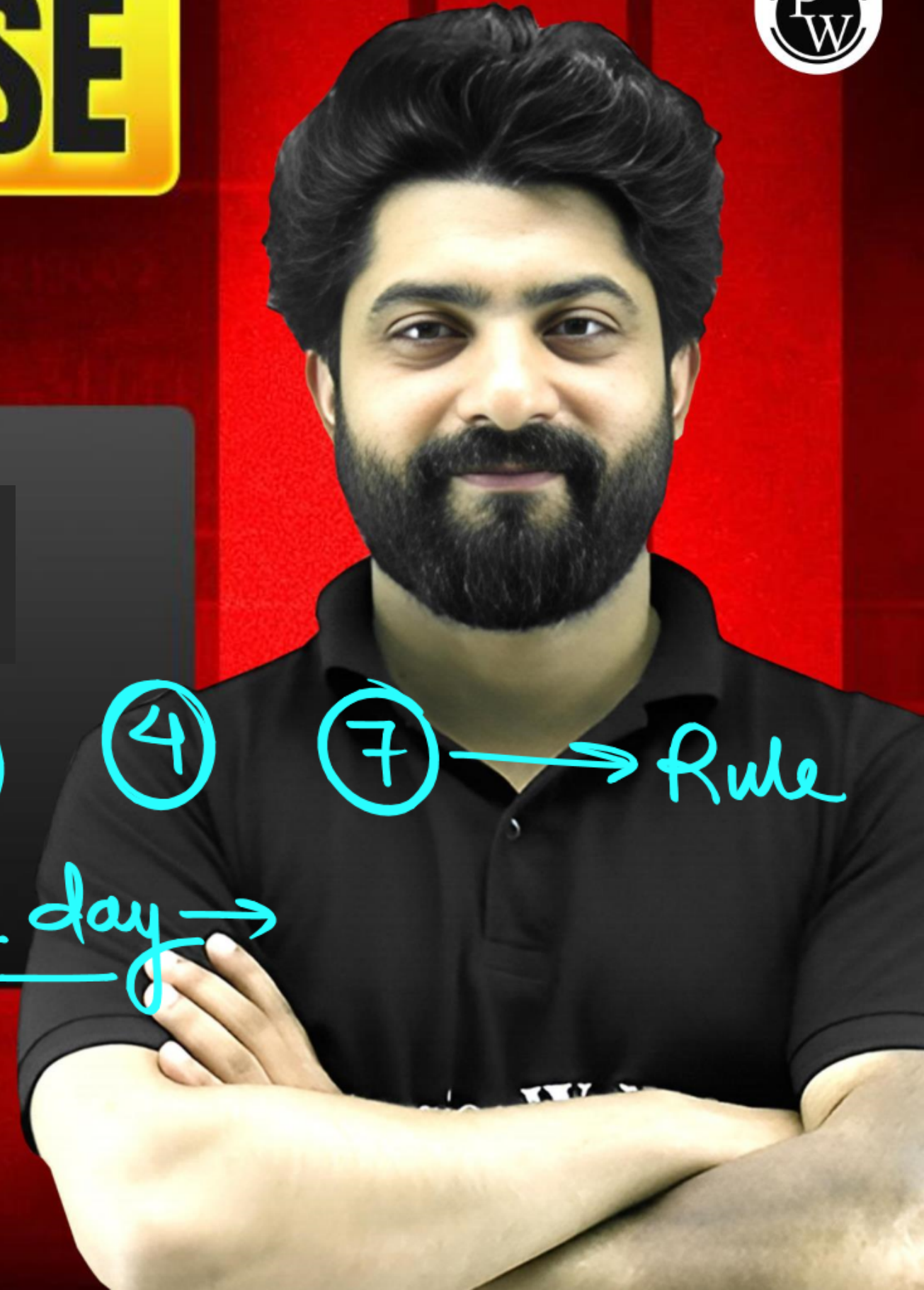
Work, Power And Energy

① ④ ⑦ → Rule
21 day →

LECTURE-2

PHYSICS

WAJAHID SIR (Gabbar Sir)



Topics To Be Covered

Revision (work. energy theorem)

- 1 Potential Energy ✓
Conservation and non-conservative force
- 2 Gravitation Potential Energy ✓
V-C - motion
- 3 Spring Potential Energy ✓
- 4 Work Done By Spring Force ✓
- 5 Law of Conservation of Energy ✓

**“KEEP GOING,
EVEN IF IT’S SLOW”**

2025/24/23/22/21/19

✓ WORK - Energy theorem

• This theorem states
Work is change in
K.E.

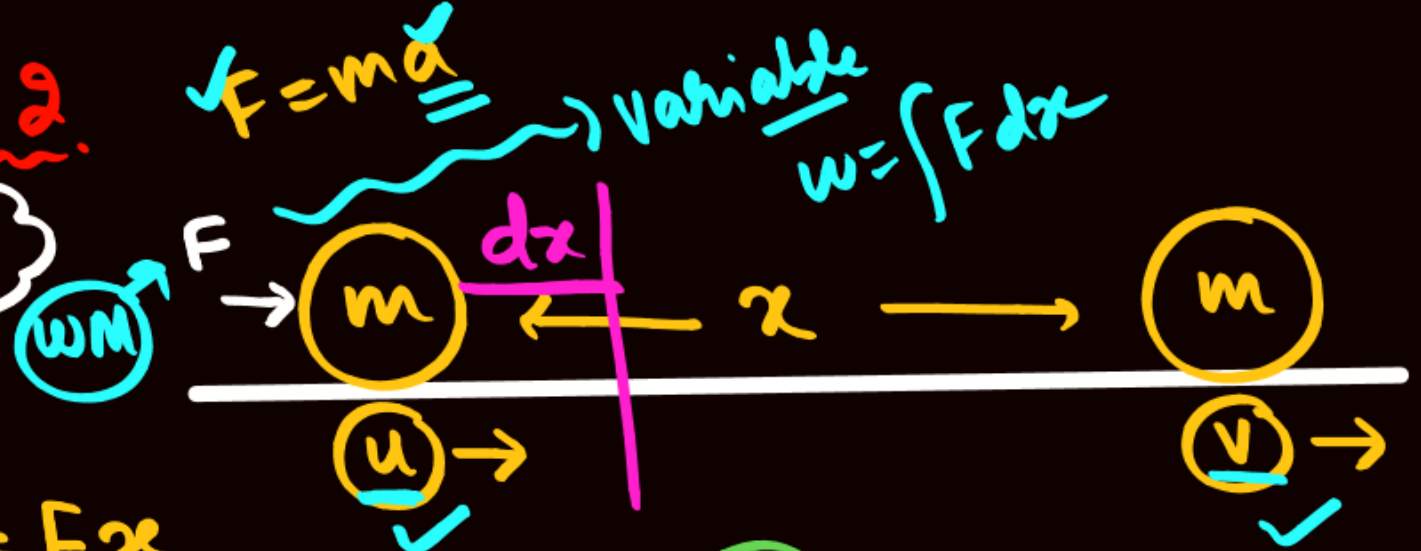
$$W = \Delta K$$

$$W = K_f - K_i$$

$$W = \frac{1}{2} m \underline{v}^2 - \frac{1}{2} m \underline{u}^2$$

lecture no. 2

Proof



$$\# \underline{W} = Fx$$

$$* \underline{dW} = F dx \quad F = ma$$

$$dW = m \underline{a} dx$$

$$dW = m v \frac{dv}{dx} dx$$

$$dW = m v dv$$

on integrating (b/s)

$$\int dW = \int m v \underline{dv}$$

$$W = \int_u^v m v dv$$

Limit (velocity) $\rightarrow u$

Note

$$a = \frac{dv}{dt}$$

$$a = \frac{dv}{dt} \times \frac{dx}{dx}$$

$$a = \frac{dv}{dx} \times \left(\frac{dx}{dt} \right)^v$$

$$a = \frac{dv}{dx} v$$

$$\underline{a} = v \frac{dv}{dx}$$

$$W = \int_u^v \underline{m} \underline{v} \underline{dv}$$

$$W = m \int_u^v \underline{v} \underline{dv}$$

$$W = m \left(\frac{v^{1+1}}{1+1} \right)_u^v$$

$$W = m \left[\frac{v^2}{2} \right]_u^v$$

$$W = m \left[\frac{v^2}{2} - \frac{u^2}{2} \right]$$

$$\int x^n dx = \frac{x^{n+1}}{n+1}$$

$$\int v^1 dv = \frac{v^{1+1}}{1+1}$$

$$\frac{x}{2} = \frac{1}{2}x$$

$$W = m \left[\frac{v^2}{2} - \frac{u^2}{2} \right]$$

$$W = \frac{mv^2}{2} - \frac{mu^2}{2}$$

$$W = \frac{1}{2}mv^2 - \frac{1}{2}mu^2$$

$$W = K_f - K_i$$

$$\boxed{W = \Delta K}$$

$$\underline{W = \Delta K \cdot \epsilon}$$

Numerical



Calculate work? JKBOSE
2016

So $K_i = \frac{1}{2} m u^2 = \frac{1}{2} \times 1 \times 10^2 = \frac{100}{2} = 50\text{ J}$

$$K_f = \frac{1}{2} m v^2 = \frac{1}{2} \times 1 \times 100^2 = \frac{10000}{2} = 5000\text{ J}$$

$$\begin{aligned} W &= K_f - K_i \\ &= 5000 - 50 \\ &= 4950\text{ J} \checkmark \end{aligned}$$

Potential Energy

P.E / U

$+q$
 $-q$

$+q$
 $-q$

Electrostatic Potential Energy $\uparrow\uparrow\uparrow$ 12th PW

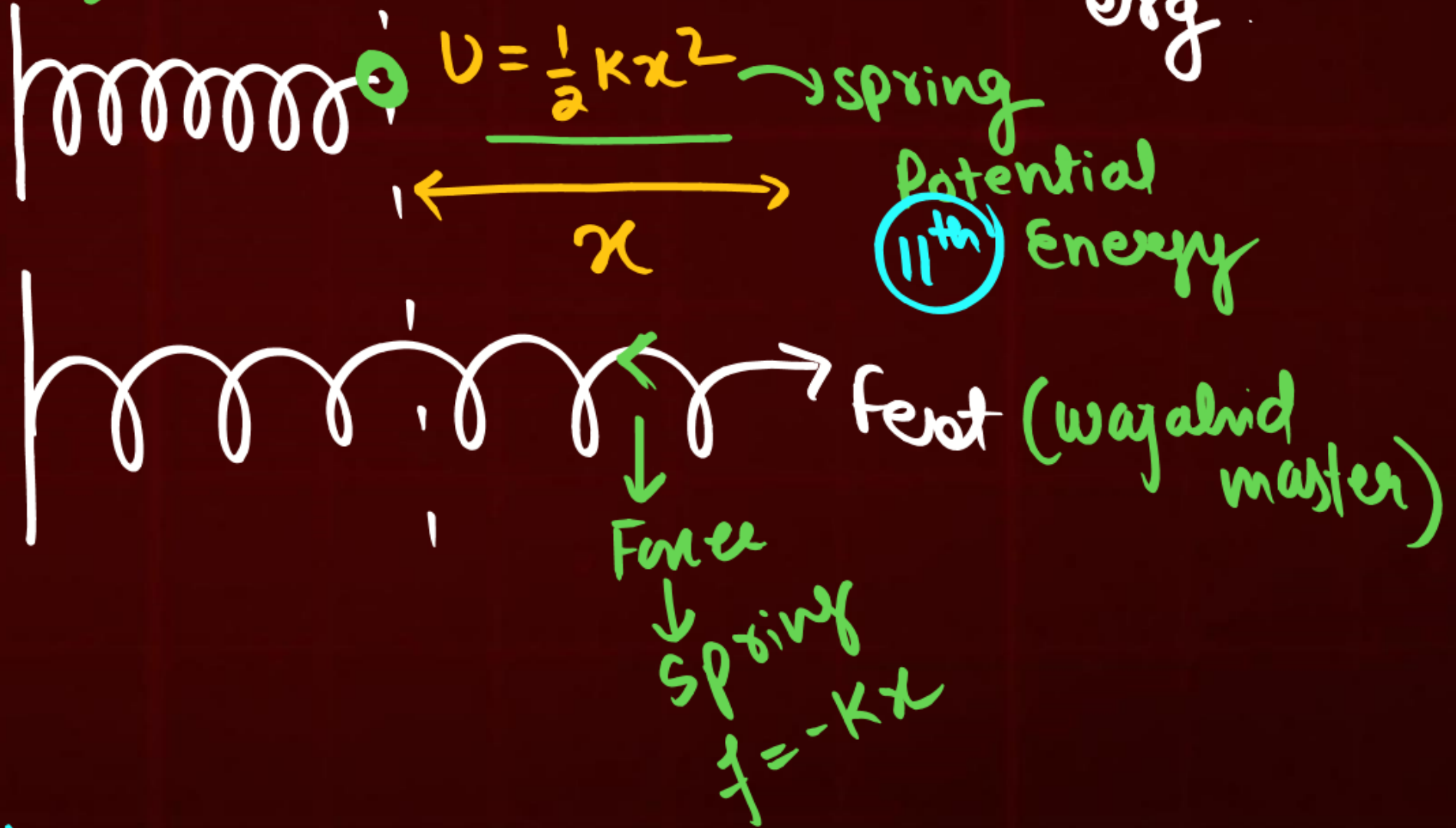
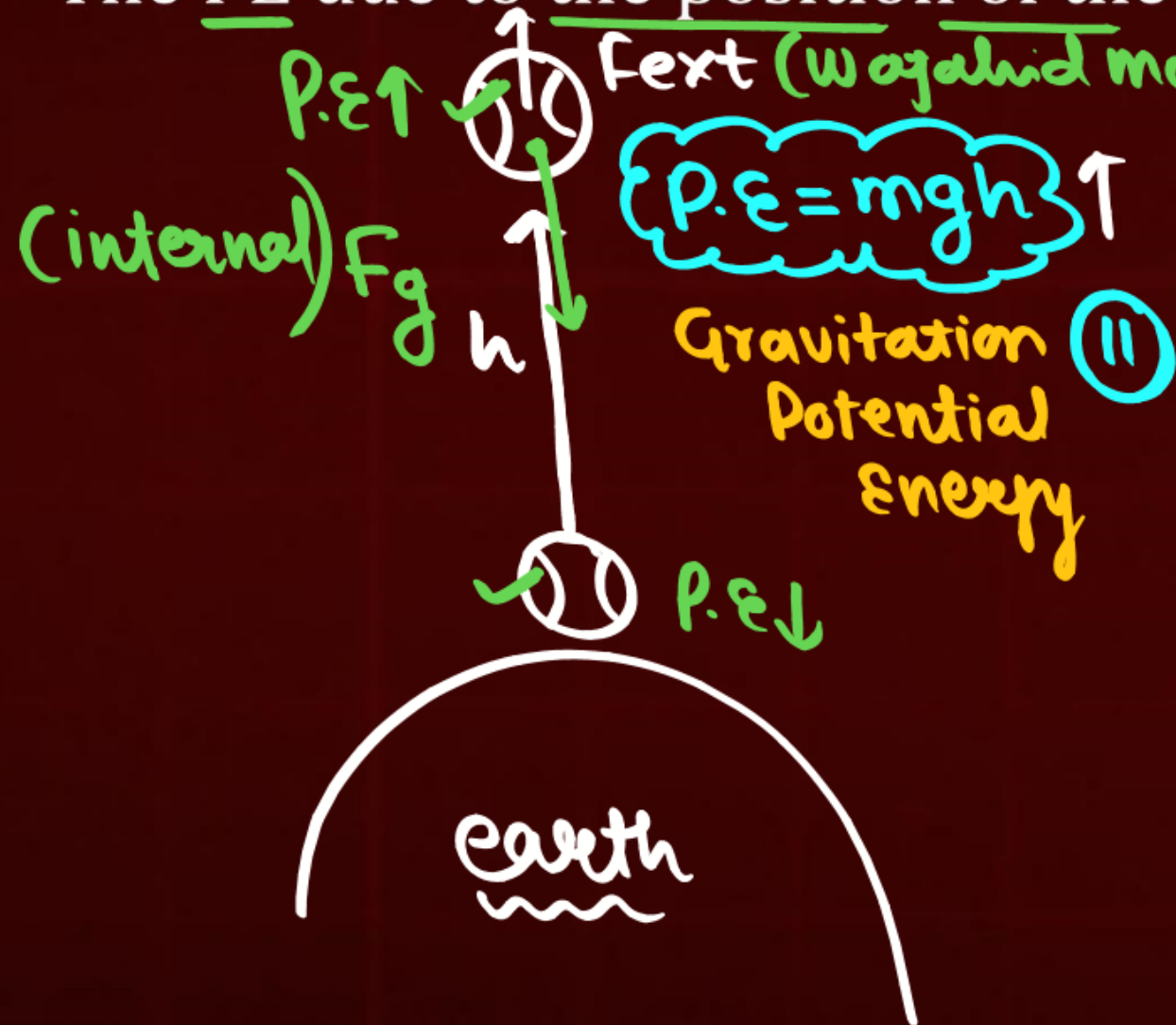
- The Energy store due to (Zabandasti) $\uparrow$ P.E $\uparrow$ or

- The PE due to the position of the particle

$$U = W_{\text{Fext}}$$

• Scalar

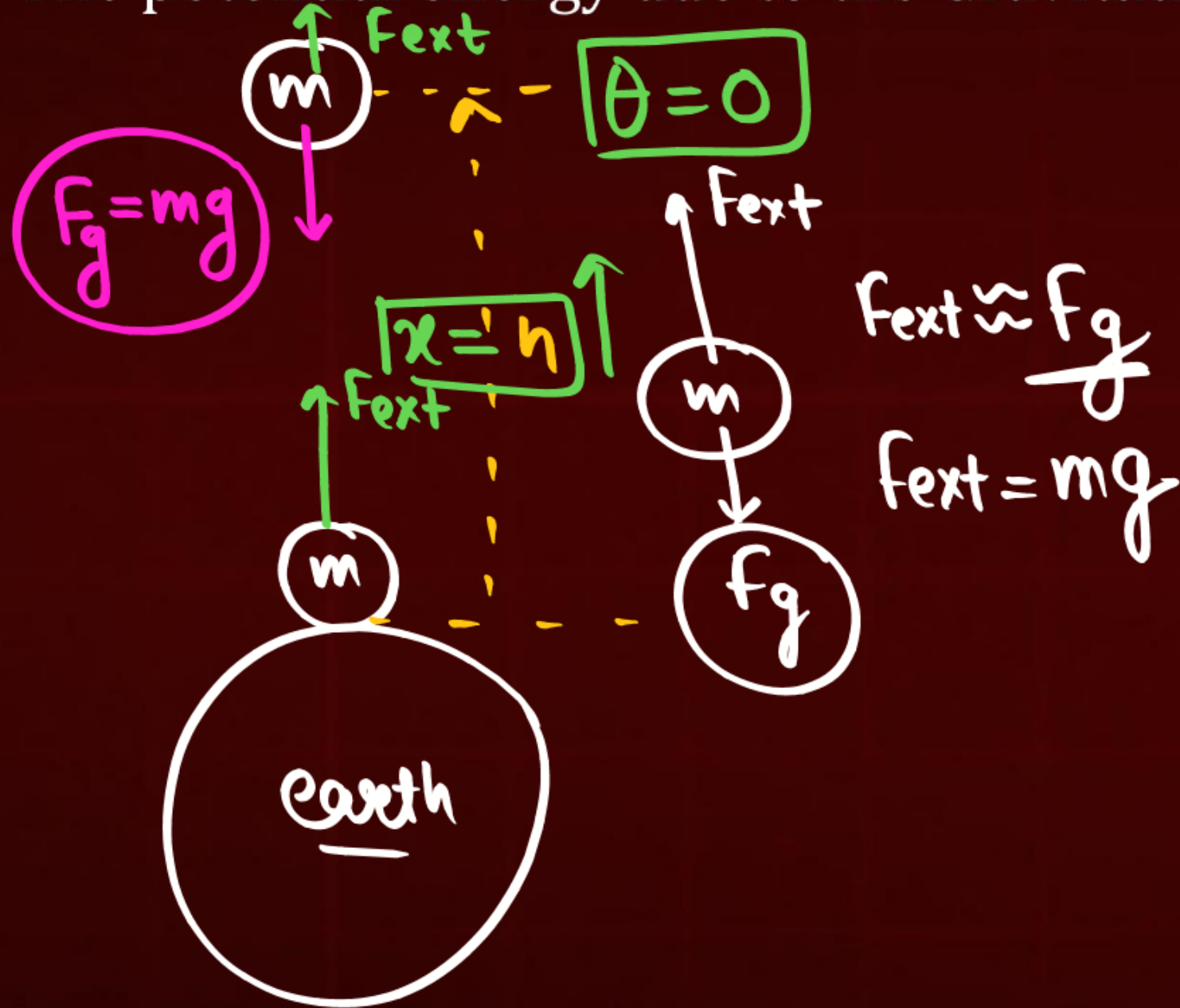
• Joule erg



① Gravitation Potential Energy



The potential energy due to the Gravitation force of the earth



$$\# U = W_{F_{ext}}$$

$$U = F_{ext} \times \cos \theta$$

$$U = \underline{F_{ext}} \times \cos 0$$

$$U = mgh$$

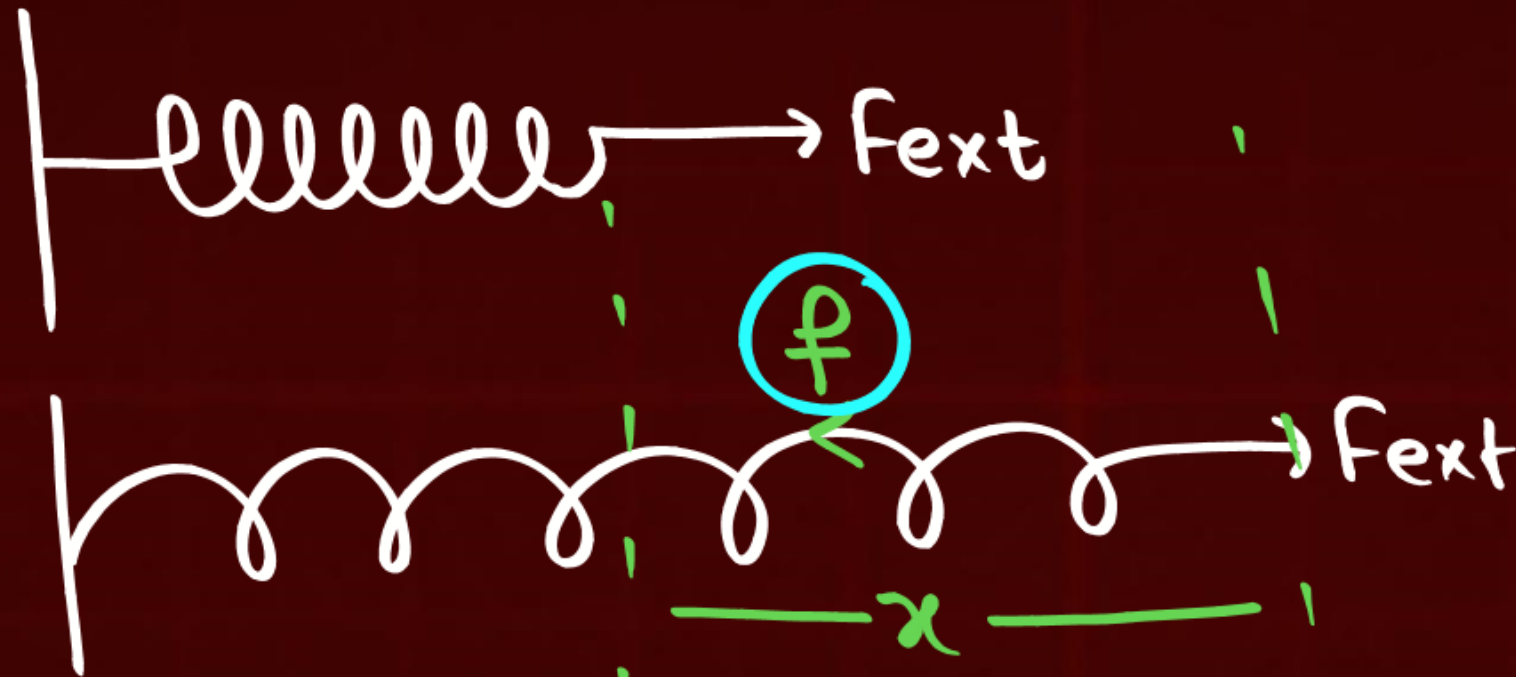
$$\uparrow U = mgh \uparrow$$

Spring Force

(NLM) ✓



The force in a spring when it is stretched or compress is the spring force



$$F = -Kx \rightarrow \text{Chapter 14}$$

$\uparrow$ $\frac{F}{K} = -x \downarrow$

$$F dx$$

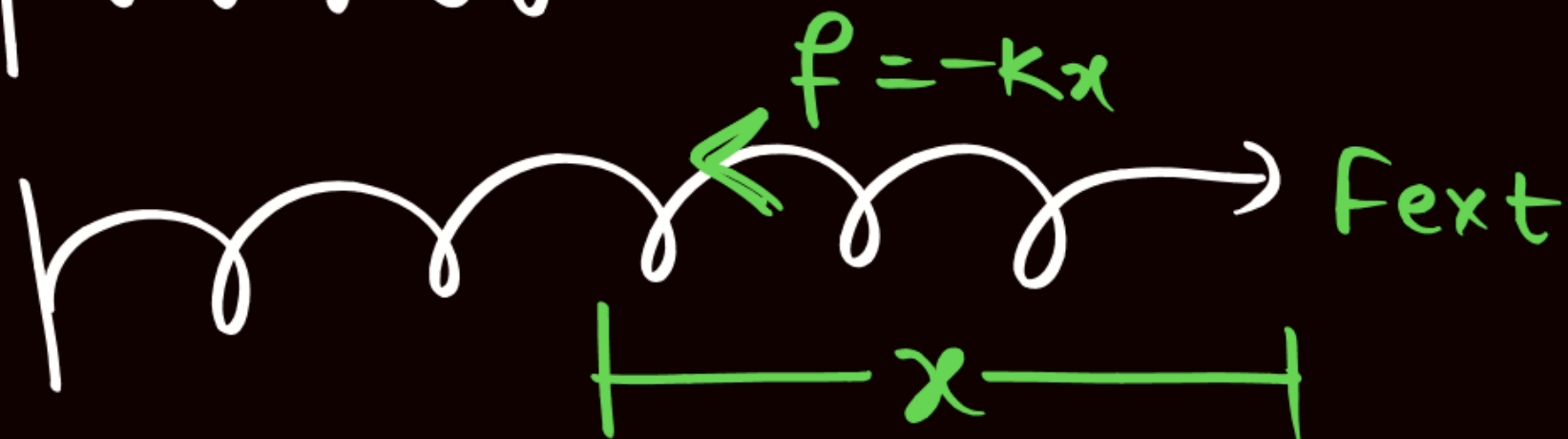
$$F = -Kx$$

$K \rightarrow$ Spring Constant

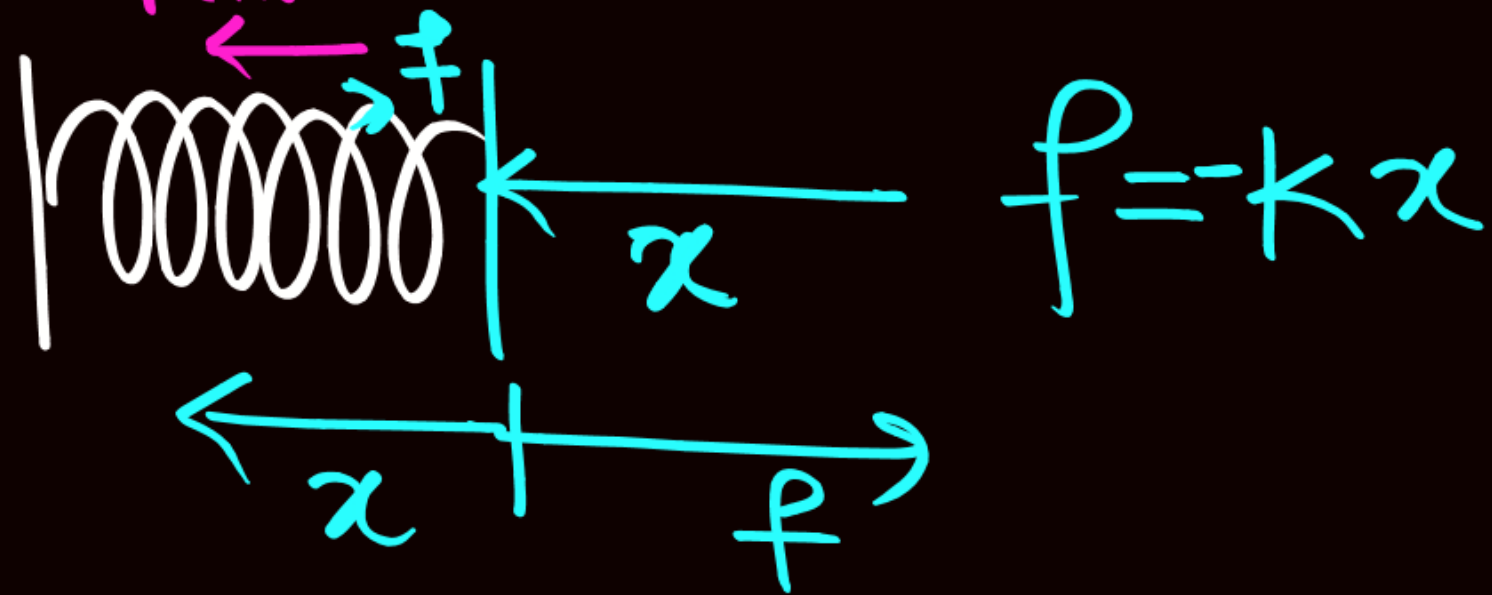
$K \downarrow$ $x \uparrow$

Spring force is in opposite direction of F_{ext} .

Stretched



Compressed



Q What is the unit of Spring Constant? (K)

Sol

$$F = -\underline{K}x$$

$$F = Kx$$

$$\frac{F}{x} = K$$

$$\frac{N}{m} = K$$

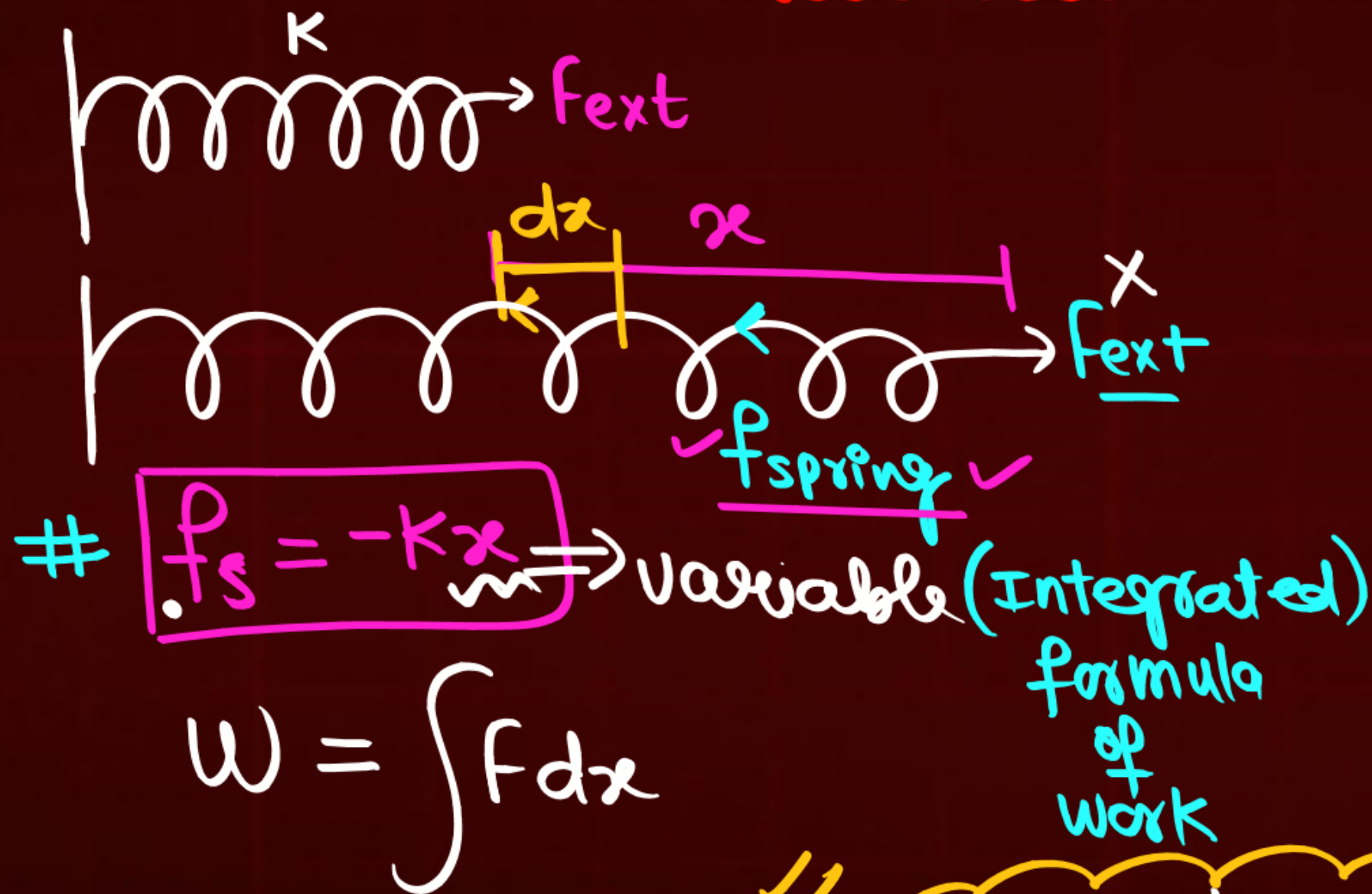
→ Spring Constant

$$\frac{N}{m} \rightarrow K$$

JKBOSE
Neet

Work Done by Spring Force

Let us consider the spring of spring constant K



$$W = -\frac{1}{2} K x^2$$

J. K. BOSE (2019 | 2018 | 2017 | 2022)



$$W = \int \underline{f} dx \quad f = -Kx$$

$$W = \int -Kx dx \quad \int x^n dx = \frac{x^{n+1}}{n+1}$$

$$W = -K \int x^1 dx$$

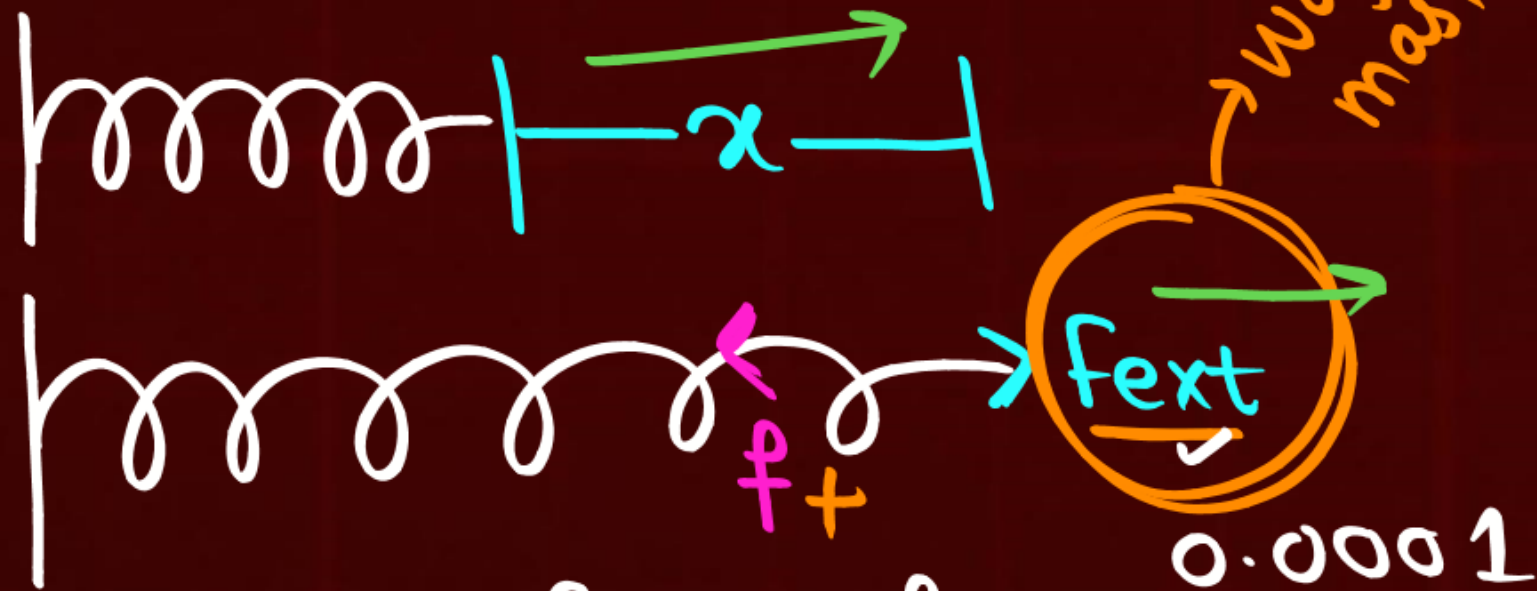
$$W = -K \frac{x^{1+1}}{1+1}$$

$$W = -\frac{Kx^2}{2} = \underline{-\frac{1}{2} Kx^2}$$

Spring Potential Energy

The potential energy store in a spring, when it is stretched

- Let us consider the spring of spring constant (K)



$$F_{ext} \approx f$$

$$F_{ext} = Kx$$

variable

$$W = \int f dx$$

$$U = W_{F_{ext}}$$

$$W_{F_{ext}} = \int F_{ext} dx$$

$$W_{F_{ext}} = \int Kx dx$$

$$W_{F_{ext}} = K \int x' dx$$

$$W_{F_{ext}} = K \frac{x^{1+1}}{1+1}$$

$$W_{F_{ext}} = K \frac{x^2}{2}$$

$$W_{\text{Fext}} = \frac{kx^2}{2}$$

$$W_{\text{Fext}} = \frac{1}{2} kx^2$$

$$U = +\frac{1}{2} kx^2$$

Law of Conservation of Energy

= 9th

- Energy always remain constant $T.E_i = T.E_f$
- Energy can neither be created nor be destroy. But it can be transfer from one form to another form.

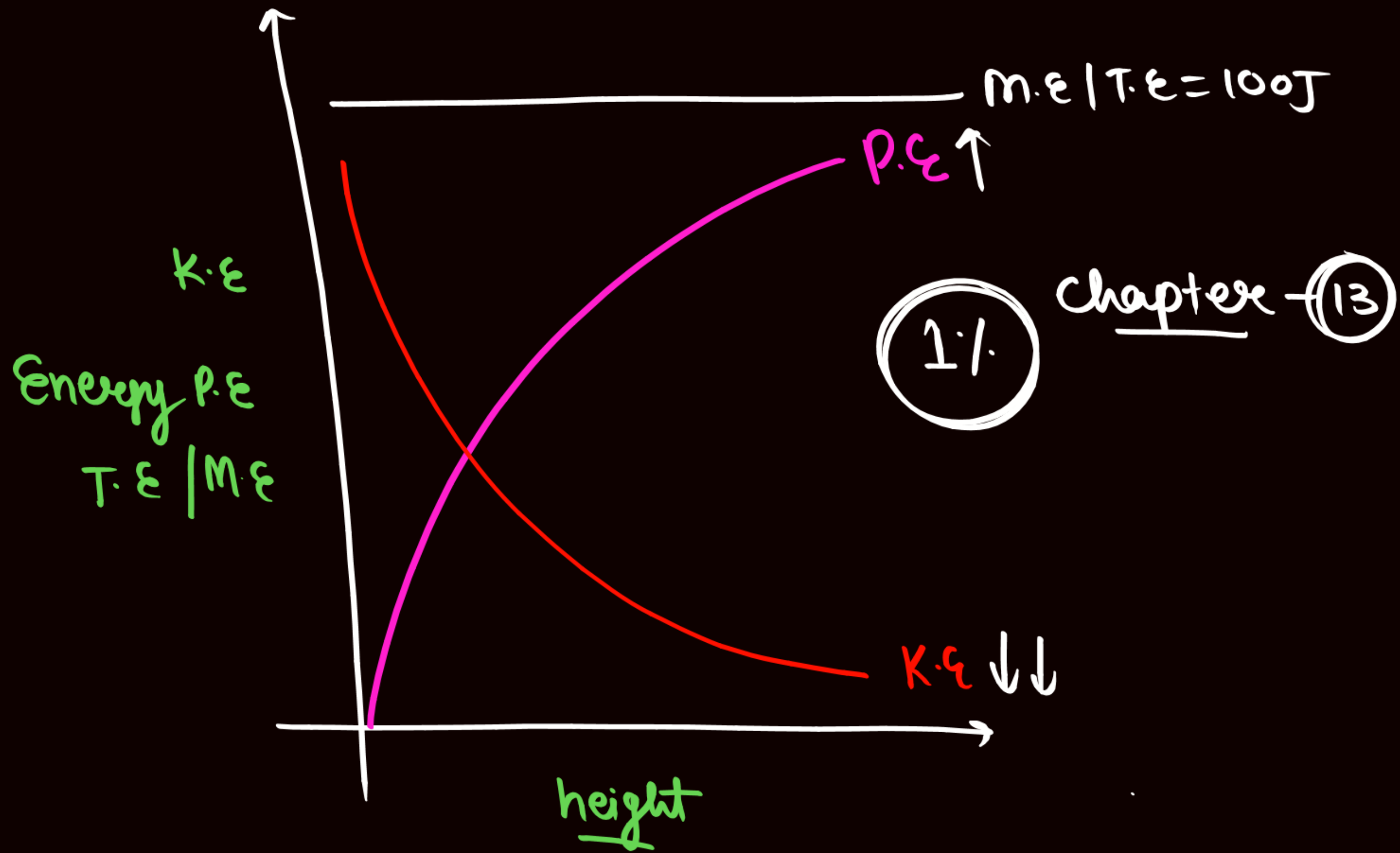
$T.E_i = T.E_f$
 $m.E_i = m.E_f$
 $100J = 100J$ } → chapter - 13

5 marks

(A) $u=0$ $K.E=0$ $P.E=100J$ $m.E = K.E + P.E = 0 + 100 = 100J$ ✓

(B) $h=1m$ $K.E=50J$ $P.E=50J$ $m.E = 50 + 50 = 100J$ ✓

(C) $K.E=100J$ $P.E=mgh=0J$ $m.E = 100 + 0 = 100J$ ✓



SUMMARY

- $W = \Delta K$ ✓
- $P.E = \frac{1}{2} kx^2$
- $W = -\frac{1}{2} kx^2$
- $P.E = mgh$
- $P.E$
- law of Conservation of Energy

HOMework

Numerical

Q



$$x = 2\text{ m}$$



$$K = 10\text{ N/m}$$

P.E Store in Spring = ?

Thank
You

